Mechanics of Human Movement

Course Progress **90.2%** (11 of 13)

- Week 11
- Week 12
  - ✓ Design Considerations: Prosthesis and Orthosis
  - ✓ Design Considerations: Prosthetic Knee
  - ✓ Journey of Standing Wheelchair Development
  - ✓ Week 12 - Lecture Notes
  - ✓ Mechanics of Human Movement : Week 12 Feedback Form
  - ✓ Quiz: Week 12: Assignment 12
- Text Transcripts
- Download Videos

Assessment

Week 12: Assignment 12

The due date for submitting this assignment has passed. Your last recorded submission was on 2026-10-08, 14:08 IST.

Due date on 2026-10-14, 23:59 IST

1. An assistive device used as a replacement for a missing limb is known as: 1 Point

○ Orthosis
○ Exoskeleton
● Prosthesis
○ Brace

Yes, the answer is correct. Score: 1 Accepted answer is Prosthesis

2. An orthosis is an assistive device used for: 1 Point

☑ Support
☐ Replacement of missing limb
☐ Communication
☑ Correction

Yes, the answer is correct. Score: 1 Accepted answer is Support, Correction

3. In a prosthetic knee joint, stability in stance can be achieved through: 1 Point

☐ Torque absorber
☑ Geometry of the joint
☐ Flexion assist
☑ Frictional braking

Yes, the answer is correct. Score: 1 Accepted answer is Geometry of the joint, Frictional braking

4. In a prosthetic knee joint, automatic adaptability to variable cadence can be achieved through: 1 Point

☐ Torque absorber
☑ Hydraulic or pneumatic units
☑ Microprocessor control
☐ Frictional control

Yes, the answer is correct. Score: 1 Accepted answer is Hydraulic or pneumatic units, Microprocessor control

5. The function of the human knee joint during walking is best approximated by a: 1 Point

● Hinge joint
○ Condyloid joint
○ Ellipsoid joint
○ Spherical joint

Yes, the answer is correct. Score: 1 Accepted answer is Hinge joint

6. Initiation of push off is comparatively easy when the load line passes _______ to the knee joint center: 1 Point

○ Anterior
○ Medial
● Posterior
○ Lateral

Yes, the answer is correct. Score: 1 Accepted answer is Posterior

7. While walking with a locked knee orthosis, the user must perform _______ to clear the ground while the orthotic leg is in swing phase: 1 Point

○ Steppage
○ Knee extension
○ Hip flexion
● Circumduction

Yes, the answer is correct. Score: 1 Accepted answer is Circumduction

8. Geometric stance stability is achieved when the load line passes _______ to the knee joint center: 1 Point

● Anterior
○ Posterior
○ Medial
○ Lateral

Yes, the answer is correct. Score: 1 Accepted answer is Anterior

9. What could be secondary consequences of prolonged sitting for wheelchair users? 1 Point

☐ Weight loss
☑ Pressure Sores
☑ Muscle contractures
☐ Increased metabolism rate

Yes, the answer is correct. Score: 1 Accepted answer is Pressure Sores, Muscle contractures

10. Which of the following are the design considerations for a prosthetic knee during swing phase? 1 Point

☐ Stabilize the knee
☑ Assist extension of the shank
☑ Assist in ground clearance
☑ Accommodate different walking speeds

Yes, the answer is correct. Score: 1

Accepted answer is Assist extension of the shank, Assist in ground clearance, Accommodate different walking speeds